Track Local : A Daily Commuter's app

1. Problem Definition

User Pain Point:
Long and unpredictable traffic congestion during metro commutes in Bangalore, leading to wasted time and frustration.

2. User Persona (Generated with AI/Prompting):

- **Name:** Rohan Sharma, 29, Software Engineer in Bangalore
- **Behavioral Insights:** Commutes daily from Whitefield to Electronic City (~20 km). Takes metro + bus, but delays and crowded trains frustrate him.
- **Emotional Drivers:** Values productivity, hates feeling stuck in traffic, wants time-efficient and cost-effective travel.
- **Current Process:** Uses Google Maps + Metro app separately, no real-time integration of delays or bus connections.
- **Opportunities:** A feature that combines **multi-modal route optimization** (metro + bus + shared cabs) with **real-time congestion updates** and **cheapest route finder**.

3. Define Product Requirements

Using **TCEPFT Framework**:

- **Task:** Help commuters choose the **fastest & cheapest metro+bus route** with real-time updates.
- **Context:** Metropolitan city commuters (Bangalore, Mumbai, Delhi) during peak hours.
- **Exemplar:** Like Google Maps, but tailored to Indian metro-bus integrations.
- **Persona:** Young working professionals (22–35), budget- and time-conscious.
- **Format:** Mobile app feature integrated into an existing commute app.
- **Tone:** Informative, simple, trustworthy.

Functional Requirements (FR):

1. User can search origin → destination and see real-time route suggestions.
2. Integration with metro/bus APIs for live status.
3. Cost comparison (e.g., Metro ₹40 + Bus ₹20 vs Cab ₹200).
4. Smart alerts (delays, strikes, overcrowding).

Non-Functional Requirements (NFR):

- Performance: Route suggestions in < 3 seconds.
- Reliability: 99% uptime.
- Security: Secure API integrations, no personal travel data leaks.

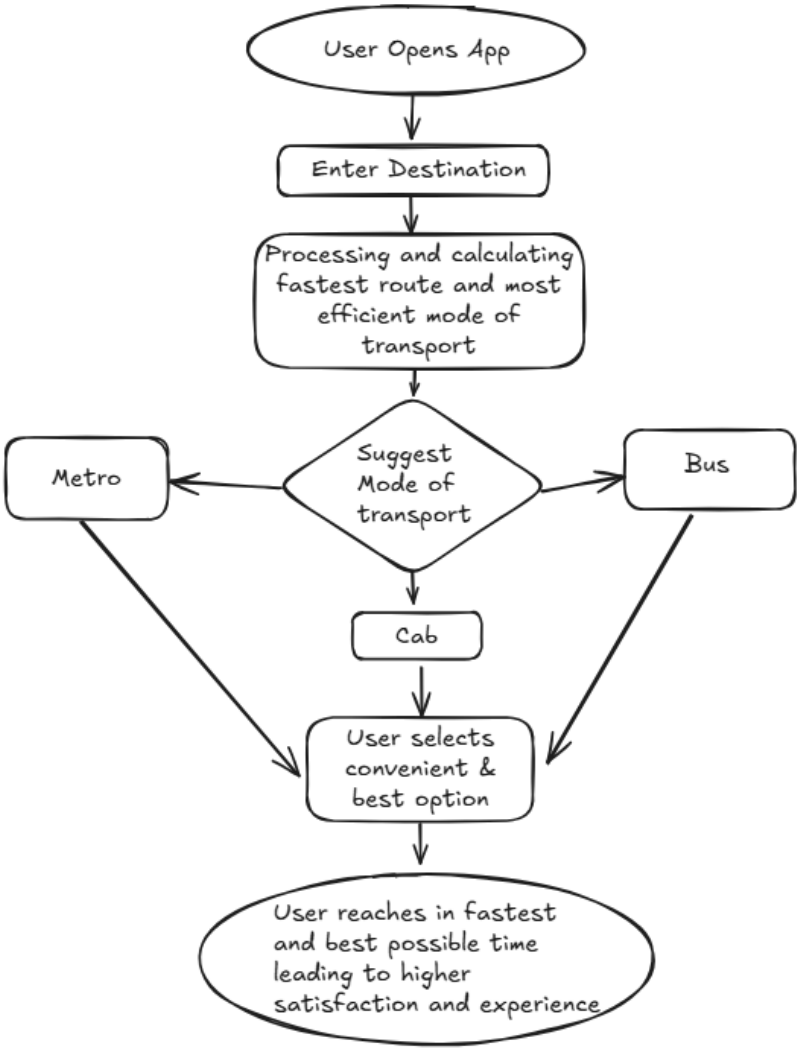
Metrics (HEART Framework):

- **Happiness:** User satisfaction ratings (target: 4.5/5).
- **Engagement:** Daily active users (DAU).
- **Adoption:** New users trying the feature in month 1.
- **Retention:** Returning users after 30 days.
- **Task Success:** % of users who reach destination using suggested routes without complaints.

4. UX Workflow & Mockups

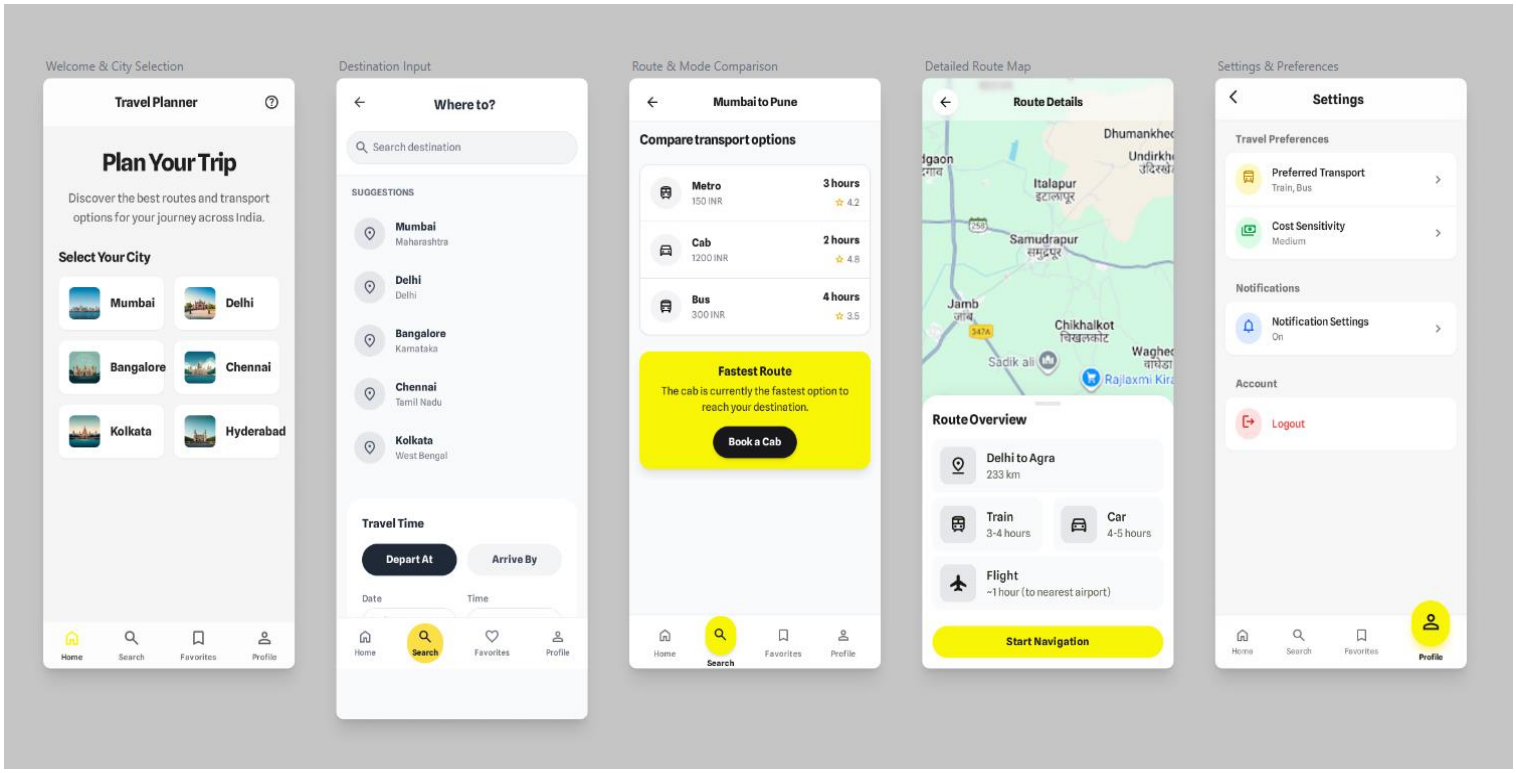
Workflow Diagram (Steps):

1. User opens app → enters start & destination.
2. AI fetches real-time metro/bus status.
3. Suggests 2–3 options: fastest, cheapest, balanced.
4. User selects → guided navigation + alerts.
5. Feedback prompt after trip (rate accuracy).



Low-Fidelity Mockup (Wireframe Idea):

- **Screen 1:** Search bar (From → To).
- **Screen 2:** Options (Fastest: 45 min, Cheapest: ₹60, Balanced).
- **Screen 3:** Live journey updates with notifications ("Metro delayed 5 min").
- **Screen 4:** Trip summary + "Rate accuracy."



Lofi UI Mockup using Google Stitch

5. Key Insights from Building the Commute Feature

1. **Commuters prioritize reliability over speed.**
   Many users are okay with slightly longer routes if they are predictable and consistent. Uncertainty in delays frustrates them more than actual time spent.
2. **Affordability is a major driver in Tier-1 and Tier-2 cities.**
   For many, the difference between ₹60 (metro+bus) and ₹200 (cab) is significant. A cheapest-route option directly addresses this pain point.
3. **Multi-modal commuting is the norm, not the exception.**
   Users combine metro, bus, autos, and walking. Current apps (Google Maps, metro apps) don't integrate these seamlessly in the Indian context.
4. **Information gap during the commute.**
   Once users start their journey, they don't get real-time alerts (e.g., "next bus delayed by 10 minutes" or "metro overcrowded"). Addressing this gap improves trust.
5. **Personalization opportunity.**
   Different users value different things: students want cheapest, office-goers want reliable, executives may want fastest. A customizable "commute preference profile" could enhance adoption.

6. Reflections from the Design Process

1. **AI as a Co-Designer:**
   Using AI to generate the **user persona** and **workflows** made the process faster and more creative. It helped explore diverse commuter scenarios that might have been overlooked manually.

2. **Prompt Engineering Matters:**
   The quality of the persona and workflows depended heavily on how prompts were framed. Asking AI to roleplay as a "commuter researcher" yielded more actionable insights.

3. **Balancing FRs vs NFRs:**
   It was easy to think of features (FRs), but harder to ensure **non-functional requirements** (speed, reliability, security). This highlighted that good product design is not just about "what it does" but "how well it does it."

4. **Metrics Guide the Design:**
   Defining **HEART metrics** early gave clarity on what success looks like (e.g., reducing commute uncertainty, improving CSAT), instead of just adding features.

5. **Limitations of AI Tools:**
   While AI sped up ideation, human judgment was still needed to validate feasibility (e.g., integration with metro/bus APIs might not be technically or politically easy).

6. **Empathy is Key:**
   The exercise reinforced that the best features come from truly understanding user frustrations—not from technology alone.

Project Submitted by Aniesh Das